Old Exam Questions
AI in Life Sciences

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How this file is organized

Questions come from two sittings: the **2024** exam (25 June 2024, Sections 1–6) and the **2026** exam (25 June 2026, Sections 7–10). The 2024 items are *multiple-select* ("which of the following is/are true" means **one or more** of the four options are correct). The 2026 items are *single-best-answer* (exactly **one** of the three options is correct). To avoid reproducing the originals verbatim, both the question wording and the answer options have been paraphrased (the meaning and the correct answers are preserved). Questions are grouped by topic rather than in their original random order. The full answer key is at the end — the question boxes themselves are intentionally left blank so the file doubles as a practice set.

Contents

Drug Discovery, Molecules & Representations

Question 1: SMILES strings

Regarding the SMILES notation for molecules, which of the following statements hold?

- ☐ SMILES is short for the *Simplified Molecular Input Line Entry System*.
- ☐ A SMILES encodes a molecule as a one-dimensional sequence of characters.
- ☐ A SMILES string cannot be turned into a molecular graph.
- ☐ Each molecule has exactly one SMILES, so no two distinct molecules can share a string.

Question 2: Reading a SMILES string

Consider the molecule *pyruvic acid* (2-oxopropanoic acid), i.e. a methyl group followed by a ketone carbonyl and then a carboxylic acid (CH_3–C(=O)–C(=O)–OH). Which of the SMILES strings below correctly encode this molecule?

- ☐ `O=C(C(=O)O)C`
- ☐ `CC(=O)C(=O)O`
- ☐ `CCCOOO`
- ☐ `OCCOO`

Question 3: Molecular fingerprints

Which of the following statements about molecular fingerprints are accurate?

- ☐ A unique molecular graph can be reconstructed from them.
- ☐ They are binary or numeric encodings of molecular structure.
- ☐ They serve similarity searches and machine-learning tasks.
- ☐ They give a 3D representation of the molecule.

Question 4: Bioassays

Select the correct statements about bioassays.

- ☐ They quantify the biological response or function that a substance triggers.
- ☐ They usually probe the substance across a range of concentrations.
- ☐ They are wet-lab procedures for characterising a compound in a biological setting.
- ☐ They are usually carried out *in vitro*, i.e. inside a living organism.

Question 5: ADME properties

Which statements about ADME (Absorption, Distribution, Metabolism, and Excretion) properties are correct?

- ☐ Lipinski's rule of five can serve as a proxy for ADME behaviour.
- ☐ Favourable ADME properties guarantee that a compound becomes an effective drug.

- ☐ Certain ADME properties, such as clearance, are hard to model.
- ☐ ADME properties are decisive for a drug's pharmacokinetics.

Question 6: Characteristics of QSAR / bioactivity data

Which of the following correctly characterise typical QSAR / bioactivity datasets?

- ☐ QSAR datasets typically contain on the order of thousands to millions of data points.
- ☐ QSAR datasets usually show a balanced mix of active and inactive compounds.
- ☐ QSAR datasets frequently include groups of structurally related compounds.
- ☐ QSAR labels are often noisy because of biological variability and experimental inconsistency.

Question 7: Applications of QSAR modeling

For which of the following applications is QSAR modeling relevant?

- ☐ Building 3D models of protein structures.
- ☐ Finding promising drug candidates.
- ☐ Estimating the toxicity of novel chemical compounds.
- ☐ Mapping the genome sequences of organisms.

Virtual Screening & QSAR Modeling

Question 8: Ligand-based virtual screening

Which statements about ligand-based virtual screening are correct?

- ☐ Molecular fingerprints are a common ingredient of ligand-based virtual screening.
- ☐ Ligand-based virtual screening cannot predict the bioactivity of untested compounds.
- ☐ Ligand-based virtual screening needs the 3D structure of the target protein.
- ☐ Ligand-based virtual screening selects compounds using bioactivity predictions alone.

Question 9: Data splits in QSAR model evaluation

When evaluating QSAR models, which of the following statements about data-splitting strategies hold?

- ☐ Temporal splits gauge performance on future, unseen data by mimicking how data accrue over time.
- ☐ A random split is generally the best evaluation scheme because it removes all bias.
- ☐ Cross-validation is seldom applied to QSAR because of its computational cost.
- ☐ Cluster splits keep similar compounds together so that structurally similar ones do not span the training and test sets.

Question 10: Early-recognition metrics in virtual screening

Because only the top of a virtual-screening ranking is followed up experimentally, recognising actives early matters most. Which of the following metrics are specifically designed to reward strong early recognition?

- ☐ Precision-Recall Curve.
- ☐ BEDROC (Boltzmann-Enhanced Discrimination of ROC).
- ☐ Average Precision (AP).
- ☐ Enrichment Factor (EF).

Question 11: Target identification / deconvolution

Which statements about target identification / deconvolution are correct?

- ☐ For a given molecule, it determines which of several simultaneously present proteins it is most likely to bind.
- ☐ It can be approached with multi-class classification.
- ☐ It can be approached with multi-task classification.
- ☐ For a given molecule, it determines every protein it would bind when several are present at once.

Question 12: Drug synergy and antagonism

Which of the following statements about drug synergy and antagonism are correct?

- ☐ Synergy means the combined effect of two or more drugs exceeds the sum of their individual effects.
- ☐ The Loewe additivity model is a standard tool for assessing synergy.
- ☐ Antagonism means the combined effect of two or more drugs exceeds the sum of their individual effects.
- ☐ Synergy/antagonism analysis applies only to chemically similar drugs.

Question 13: Random Forests vs. Graph Neural Networks

Comparing Random Forests (RF) and Graph Neural Networks (GNNs) for molecular modeling, which statements are correct?

- ☐ GNNs are inherently superior for QSAR because they capture the graph structure of molecules, giving more accurate predictions in every case.
- ☐ RF models overfit less than GNNs, which makes them more reliable on datasets with many features but few samples.
- ☐ RF models are usually more interpretable than GNNs, so the contribution of individual molecular features is easier to read off.
- ☐ GNNs are less sensitive to input-data quality and variability than RFs, making them more consistent across different datasets.

Graph Neural Networks & Advanced Methods

Question 14: Molecular graphs and graph neural networks

Which statements about molecular graphs and graph neural networks are correct?

- ☐ A molecular graph can be stored as two matrices: one for the node features and one for the edges between nodes.
- ☐ Node features may include the atomic number, hybridization state, and formal charge.
- ☐ A GNN uses only the node features to form its final prediction.
- ☐ Bond-type information (e.g. single vs. double bond) cannot be represented in a GNN.

Question 15: Limitations of GNNs in molecular modeling

Which of the following are genuine limitations of GNNs in the context of molecular modeling?

- ☐ GNNs cannot predict any molecular property with accuracy.
- ☐ GNNs may struggle to capture long-range interactions between atoms.
- ☐ GNNs can have trouble with very large, complex molecular graphs.
- ☐ GNNs need considerable computational resources to train.

Question 16: Multi-modal contrastive learning

Which statements about multi-modal contrastive learning are correct?

- ☐ Multi-modal contrastive learning embeds several modalities (e.g. molecules and proteins) into one shared latent space.
- ☐ Multi-modal contrastive learning is a self-supervised method.
- ☐ Multi-modal contrastive learning concentrates the (latent) representations of several modalities (e.g. molecules and proteins) to make predictions.
- ☐ Multi-modal contrastive learning is an unsupervised method.

Question 17: Multiple instance learning (MIL)

Which of the following statements about multiple instance learning (MIL) hold?

- ☐ MIL trains on sets (bags) of instances in which each instance carries its own label.
- ☐ MIL needs the bag size to be fixed.
- ☐ MIL trains on sets (bags) of instances that share a single label.
- ☐ A MIL model should be invariant to the ordering of the instances.

Generative Models & Chemical Synthesis

Question 18: Generative models for small molecules

Which statements about generative models for small molecules are correct?

- ☐ Because SMILES syntax is intricate with long-range dependencies, SMILES generators yield fewer than 10% valid strings.
- ☐ Generative models are used to explore the already-synthesised, known chemical space.
- ☐ Generative adversarial networks cannot generate small molecules.
- ☐ Autoregressive models can generate SMILES representations of small molecules.

Question 19: Goal-directed molecule generation

Which of the following correctly describe goal-directed molecule generation?

- ☐ Goal-directed generation is concerned only with reproducing existing structures that already have the desired properties.
- ☐ Goal-directed generation aims to produce molecules whose properties resemble those of the training set.
- ☐ Reinforcement learning can be used to drive goal-directed molecule generation.
- ☐ Goal-directed generation aims to produce molecules with particular, pre-specified properties.

Question 20: Fréchet ChemNet Distance (FCD)

Which statements about the Fréchet ChemNet Distance (FCD) and how it is computed are correct?

- ☐ FCD measures the distance between the individual atoms of molecules.
- ☐ FCD compares the means and covariances of the molecules' latent representations.
- ☐ FCD is computed straight from the raw molecular structures with no transformation.
- ☐ FCD is a metric comparing the distribution of generated molecules with that of real ones.

Question 21: AI for chemical reactions

Which of the following statements about machine learning for chemical reactions are correct?

- ☐ Some approaches use so-called reaction templates, i.e. transformation rules for molecules.
- ☐ Deep-learning reaction-outcome predictors can be trained on databases of known reactions.
- ☐ Transformer architectures have been applied to predicting reaction outcomes.
- ☐ Predicting reactions with machine learning requires quantum-mechanical data because of bond changes and electron movement.

Question 22: Synthesis planning

Which statements about synthesis planning are correct?

- ☐ Synthesis planning designs a sequence of chemical reactions that yields a target compound.
- ☐ Retrosynthetic analysis is central to it, decomposing complex molecules into simpler starting materials.
- ☐ Modern deep-learning methods have made synthesis planning trivial.
- ☐ Synthesis planning amounts to nothing more than choosing the reactants.

Question 23: Retrosynthesis as a reinforcement-learning game

Multi-step retrosynthesis can be cast as a game solvable by reinforcement learning. Which of the following statements about this "retrosynthesis game" are correct?

- ☐ In the "game", an action consists of choosing reactants for a reaction.
- ☐ In the "game", the state is the set of available building blocks.
- ☐ In the "game", the environment carries out the reaction and returns the new molecules.
- ☐ In the "game", the reward is granted once the product has been synthesised from the building blocks.

Medical Data & Imaging

Question 24: Tabular medical data: types and missing values

Suppose we are given a table of patient medical records with three columns: "smoking status" (one of three string values: non-smoker, former smoker, current smoker), "cholesterol level" (floating-point measurements in mg/dL), and "diabetes risk score" (integer values in $[0, 10]$ indicating the likelihood of developing diabetes). Which of the following statements are correct?

- ☐ Missing "cholesterol level" entries can be filled with the column's median to cope with skew.
- ☐ The "smoking status" column is continuous, since it reflects patients' various smoking habits.
- ☐ "cholesterol level" is ordinal, because the measurements can be placed in increasing order.
- ☐ Missing "smoking status" entries can be filled with the column's mean to cope with skew.

Question 25: Batch effects in medical data

Which statements about batch effects in medical data are correct?

- ☐ Overlooking batch effects can yield models that separate batches instead of the true biological conditions.
- ☐ Batch effects vanish simply by collecting more samples.
- ☐ Batch effects introduce artificial differences that stem from varying experimental conditions.

- ☐ Batch effects are generally helpful, supplying needed variability to the data.

Question 26: Domain shifts in medical and healthcare data

Medical and healthcare data are frequently subject to domain shifts. Using the lecture notation (y = disease status, e.g. "healthy"/"diseased"; x = patient features, e.g. age, blood parameters, heart rate), which of the following statements about domain shifts are correct?

- ☐ A model with high predictive quality on a randomly drawn validation set will, by that fact alone, adapt to domain shifts and keep its quality over time.
- ☐ The patient-feature distribution $p(x)$ is never touched by domain shifts.
- ☐ During waves of an infectious disease (such as COVID-19), the prevalence—and thus the probability $p(y)$ of observing the phenotype—changes.
- ☐ If the way the status y is diagnosed changes, e.g. an antigen test is replaced by a PCR test, this is a concept shift $p(y \mid x)$.

Question 27: Medical imaging modalities

Which statements about medical imaging modalities are correct?

- ☐ Computed Tomography (CT) builds its body images from ultrasound.
- ☐ X-rays distinguish tissues of differing density, such as bone versus soft tissue.
- ☐ Magnetic Resonance Imaging (MRI) yields detailed images of internal structures using strong magnetic fields and radio waves.
- ☐ Optical coherence tomography forms images of organs and tissues using multiple X-rays.

Question 28: Most common task in medical imaging

Compared with typical computer-vision tasks, which task is especially prominent in medical imaging?

- ☐ Image generation.
- ☐ Classification.
- ☐ Segmentation.
- ☐ Object detection.

Question 29: The U-Net architecture

Which of the following correctly describe the U-Net architecture?

- ☐ U-Net was the first vision model whose encoder relied on self-attention rather than convolutions.
- ☐ The U-Net architecture has a symmetric encoder and decoder.
- ☐ U-Net was created specifically for the object-detection task.
- ☐ To help decoding, encoder feature maps are concatenated with those of the matching up-sampling level of the decoder.

Question 30: The "Clever Hans" effect

In the context of AI for medical imaging, what does the "Clever Hans" effect refer to?

- ☐ A bias that arises when up-scaling the resolution of low-quality medical images.
- ☐ A case where a model looks like it performs well but is actually leaning on irrelevant features or artifacts in the data.
- ☐ A case where an AI predicts outcomes incorrectly because its training data were biased.
- ☐ The use of data augmentation to make medical-imaging models more robust.

Question 31: Explainable AI and interpretability in medical imaging

Which statements about explainable AI and interpretability in medical imaging are correct?

- ☐ Gradient integration propagates a relevance value backwards through the layers to the input, estimating each input feature's relevance.
- ☐ Explainable-AI methods such as Grad-CAM help visualise which parts of a medical image matter most for a prediction.
- ☐ Occlusion analysis deletes patches from the training images to make the model more interpretable.
- ☐ Counterfactual explanations generate alternative images showing what would have had to change for a different diagnosis.

Question 32: AlphaFold 2

Which of the following statements about AlphaFold 2 are correct?

- ☐ AlphaFold 2 relies on quantum-mechanical calculations to sharpen its predictions.
- ☐ AlphaFold 2 draws on multiple sequence alignments and homologous protein structures to improve its accuracy.
- ☐ AlphaFold 2 is a deep-learning model that predicts a protein's three-dimensional structure from its amino-acid sequence.
- ☐ AlphaFold 2 is built on a convolutional neural network architecture.

Biological Sequences

Question 33: Composition of biological sequences

Which statements about the composition of biological sequences are correct?

- ☐ Protein sequences are chains of amino acids joined by peptide bonds.
- ☐ DNA sequences are built from nucleic acids.
- ☐ Biological sequences obey no particular patterns or rules in their make-up.
- ☐ RNA sequences are built from lipids.

Question 34: Deep learning for biological sequences

Which of the following statements about deep learning methods for biological sequences are correct?

- ☐ Feature extraction from biological sequences can reveal motifs, i.e. recurring patterns or elements within them.
- ☐ One-hot encoding is a common way to prepare biological sequences as input for deep models.
- ☐ Deep-learning techniques apply to tasks such as classification, segmentation, and generative modelling of biological sequences.
- ☐ 1D convolutions, Recurrent Neural Networks (RNNs), Long Short-Term Memory networks (LSTMs), and Transformers are all used to analyse biological sequences.

2026 exam — single-best-answer

The remaining sections are from the **2026** exam (25 June 2026). Each of these questions has **exactly one** correct option among the three listed. Both the question and the options have been paraphrased; the meaning and correct answers are preserved.

Molecules, Representations & Cheminformatics (2026)

Question 35: Molecular graphs vs. fingerprints

Which of the following statements accurately describes the relationship between molecular graphs and molecular fingerprints?

- ☐ A fingerprint can be computed from a molecular graph, yet the exact graph usually cannot be recovered from the fingerprint.
- ☐ Fingerprints and molecular graphs must hold the same structural information.
- ☐ A molecular graph can always be reconstructed exactly from its fingerprint.

Question 36: Converting between SMILES and fingerprints

Using standard cheminformatics tools, in which direction is conversion between a SMILES string and a molecular fingerprint actually possible?

- ☐ Cheminformatics tools can convert SMILES and fingerprints into one another in both directions.
- ☐ Cheminformatics tools can turn a SMILES into a fingerprint, but not the other way round.
- ☐ Cheminformatics tools can turn a fingerprint into a SMILES, but not the other way round.

Question 37: Tanimoto coefficient

On what quantity is the Tanimoto coefficient based when it scores how similar two molecular fingerprints are?

- ☐ It takes the count of shared features over the total number of features present in either molecule.
- ☐ It superimposes three-dimensional pharmacophore points and averages their spatial distances.
- ☐ It compares the core scaffolds, scoring one whenever the scaffolds coincide.

Question 38: Aspirin SMILES

Which of the following SMILES strings represents aspirin (acetylsalicylic acid)?

- ☐ `O=C(C)Oc1ccccc1C(=O)O`
- ☐ `cc(=O)Oc1ccccc1c(=O)O`
- ☐ `O=C[C]Oc1ccccc1C[=O]O`

Question 39: Bioactivity datasets

Which of the following best captures what bioactivity datasets usually look like in practice?

- ☐ They are usually large, balanced, and almost free of measurement noise.
- ☐ They are frequently small, imbalanced, and noisy, with far more inactive than active compounds.
- ☐ They are cheap to produce and therefore cover most compound–assay combinations.

Question 40: The Design–Make–Test–Analyse cycle

In drug discovery, which option correctly captures how the Design–Make–Test–Analyse (DMTA) cycle proceeds?

- ☐ Compounds are tested before being designed, and only the active ones are afterwards synthesized.
- ☐ Compounds are designed and made just once, after which testing removes any need for further design.
- ☐ Compounds are designed, made, and tested experimentally, and the analysis steers the next design round.

Virtual Screening, QSAR & Targets (2026)

Question 41: Structure-based vs. ligand-based drug design

What is the essential difference between structure-based drug design (SBDD) and ligand-based drug design (LBDD)?

- ☐ SBDD needs known active ligands, whereas LBDD needs the target's 3D structure.
- ☐ Both SBDD and LBDD need a high-resolution 3D structure of the target.
- ☐ SBDD relies on the target's 3D structure, whereas LBDD can work from known ligands without that structure.

Question 42: Similarity search vs. model-based search

In what way do similarity search and model-based search differ from each other?

- ☐ Similarity search is ligand-based, whereas model-based search must have the 3D structure of the target's binding site.
- ☐ Similarity search retrieves molecules resembling a query, whereas model-based search predicts activity with a model trained on labelled data.
- ☐ Similarity search learns a decision function from labelled actives and inactives, whereas model-based search ranks compounds by closeness to a query.

Question 43: Early-recognition metrics (odd one out)

Of the metrics below, which one is *not* built to reward finding actives early, i.e. near the top of a ranked virtual-screening list?

- ☐ BEDROC.
- ☐ AUC-ROC.
- ☐ Enrichment factor.

Question 44: Evaluating a virtual-screening model

A virtual-screening model orders a million compounds, yet your budget only permits assaying the top 1%. Which model-selection strategy best fits this situation?

- ☐ Pick the model with the best accuracy at a 0.5 probability threshold, since ranking above that threshold no longer matters once compounds are classified.
- ☐ Pick the model with the highest overall AUC-ROC, since weighting all ranking positions equally guarantees the best enrichment in the top 1%.
- ☐ Favour an early-recognition metric such as the enrichment factor near the 1% cutoff or BEDROC, since the top of the ranking decides experimental usefulness.

Question 45: Defensible generalization estimates in QSAR

In a QSAR pipeline, which procedure yields the most trustworthy estimate of how well the model generalises?

- ☐ Fix the split first, carry out feature and hyperparameter selection on the training data, and reserve the held-out test set for the final evaluation only.
- ☐ Split first, then compare models and hyperparameters on the test set and report the best test-set score.
- ☐ Select features on the whole dataset, then split the reduced data and tune hyperparameters on the training part only.

Question 46: Compound-series bias

What does compound-series bias refer to, and why does it undermine QSAR model evaluation?

- ☐ Because many compounds are closely related, a random split can put near-analogues in both training and test, underestimating the generalization error.

- ☐ Test-compound information leaks into feature or hyperparameter selection, biasing the reported score even when compounds are structurally unrelated.
- ☐ The active class is far smaller than the inactive one, so the majority class dominates performance even when the chemical series are diverse.

Question 47: Drug synergy and model symmetry

Which option combines a correct definition of drug synergy with a sensible property that a model predicting it should have?

- ☐ The combined effect is larger than the additive one, and the prediction should stay the same when the two compounds are swapped.
- ☐ The combined effect equals the additive expectation, and the prediction may differ when the pair is supplied in reverse order.
- ☐ The combined effect is below the additive expectation, and the prediction should be unaffected by the order of the compounds.

Question 48: Proteochemometric modelling

In what respect does proteochemometric modelling go beyond classical single-target QSAR?

- ☐ It uses representations of both the molecule and the protein target together.
- ☐ It uses molecular representations with a softmax over the candidate protein targets.
- ☐ It uses molecular representations with one separate prediction head per protein target.

Question 49: Target deconvolution / target fishing

How is the task known as target deconvolution (or target fishing) defined?

- ☐ Given one protein and a compound library, rank every molecule by predicted affinity to that protein.
- ☐ Given one molecule and several candidate proteins, predict which protein it is most likely to bind.
- ☐ Given one molecule, predict binding to each candidate protein independently, so any number of targets may be assigned at once.

Graph Neural Networks (2026)

Question 50: Reordering atoms in a molecular graph

Suppose the atoms of a molecular graph are given new indices while the bonds between them are left unchanged. What behaviour do we expect from a graph neural network?

- ☐ The node representations are permuted accordingly, but the graph-level prediction may change because the adjacency matrix was reordered.
- ☐ Both the node representations and the graph-level prediction stay in exactly the same tensor positions.

- ☐ The node representations are permuted accordingly, while the graph-level prediction stays unchanged.

Question 51: Steps of an MPNN layer

What is the usual order of operations carried out inside a single Message Passing Neural Network (MPNN) layer?

- ☐ Generate messages, aggregate them, then update.
- ☐ Generate messages, apply graph convolution, then aggregate.
- ☐ Apply graph convolution, generate messages, then update.

Question 52: Oversmoothing in GNNs

Once a large number of message-passing layers are stacked, the node embeddings across a molecular graph end up nearly identical. Why does this cause trouble?

- ☐ Long-range information is squeezed through narrow graph bottlenecks, so distant signals cannot travel effectively.
- ☐ The model may no longer tell local atom environments apart because the node embeddings grow too similar.
- ☐ Important information never reaches distant nodes because too few message-passing steps were taken.

Question 53: Contrastive learning

By what mechanism does contrastive learning build a joint molecule–target embedding space?

- ☐ It clusters molecules and targets separately and assumes equal-index clusters are binding pairs.
- ☐ It pulls associated molecule-target pairs closer together and pushes mismatched pairs apart.
- ☐ It concatenates molecule and target representations and fits a supervised activity regressor without comparing pairs in a shared space.

Question 54: Multiple Instance Learning

How does Multiple Instance Learning depart from ordinary supervised learning, in which every feature vector carries a single label $(\boldsymbol{x}, y)$?

- ☐ …we are given a bag of feature vectors that share a single label $(\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N\}, y)$.
- ☐ …we are given a bag of feature vectors paired with a bag of labels $(\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N\}, \{y_1, \ldots, y_M\})$.
- ☐ …we are given single feature vectors paired with a bag of labels $(\boldsymbol{x}, \{y_1, \ldots, y_M\})$.

Generative Models & Synthesis Planning (2026)

Question 55: Reward hacking a QSAR proxy

A generator is trained to maximise a QSAR-based activity score and learns to output molecules the score rates as highly active, yet later wet-lab assays find them inactive. What most plausibly went wrong?

- ☐ The activity score was supplied only after each molecule was finished, so the reward was delayed.
- ☐ The generator gamed weaknesses in the proxy scoring model instead of raising genuine experimental activity.
- ☐ The generator made too few structurally distinct molecules, lowering the diversity of candidates.

Question 56: Combating analogue rediscovery (odd one out)

A goal-directed generator keeps producing close analogues of molecules it has already scored highly. Which of the following would *not* help to counteract this tendency?

- ☐ Blend samples from a fixed exploratory generator with those of the trainable generator while generating SMILES.
- ☐ Penalise the reward of molecules resembling earlier high-scoring ones kept in a memory buffer.
- ☐ Push exploitation harder by always taking the highest-scoring molecule as the next training example.

Question 57: Autoregressive molecular reinforcement learning

When molecule construction is performed autoregressively and treated as a reinforcement-learning problem, which assignment of the RL roles is correct?

- ☐ The partial molecule is the state and the next building step is the action, but the scoring function is the policy and the generator gives the reward.
- ☐ The finished molecule is the state, the scoring function is the action, and each partial molecule gets its own final reward.
- ☐ The partial molecule is the state, the next building step is the action, the generator is the policy, and the finished molecule earns the reward.

Question 58: Diffusion vs. one-shot latent decoding

What sets diffusion-based molecular generation apart from decoding a molecule from a single latent vector in one pass?

- ☐ It maps a single latent vector straight to a finished molecule in one decoder pass.
- ☐ It begins from noise and repeatedly applies a learned denoising process.
- ☐ It builds the molecule step by step, predicting the next token or graph action from the partial molecule.

Question 59: Conditional generation

Trained on molecule–logP pairs, a model takes a target logP value as input and emits molecules in a single forward pass, never querying a logP scoring function during generation. Which generative paradigm is this?

- ☐ Conditional generation.
- ☐ Goal-directed generation.
- ☐ Distribution learning followed by unconditional sampling.

Question 60: Distribution learning

In molecular generation, which option most accurately captures what distribution learning is?

- ☐ Distribution learning produces molecules conditioned on side information such as protein targets or desired property values.
- ☐ Distribution learning seeks to produce molecules resembling the training set by learning its underlying distribution.
- ☐ Distribution learning concentrates on producing molecules optimised for specific biological or physicochemical properties.

Question 61: Mode collapse

In a generative model for molecules, what does the term mode collapse refer to?

- ☐ The model produces molecules that are valid and novel but match the training-set property distribution poorly.
- ☐ The model keeps emitting the same or nearly the same molecules.
- ☐ The model emits almost only molecules that already appear in the training set.

Question 62: Reaction prediction and retrosynthesis

Which of the options correctly pairs each task with its corresponding inputs and outputs?

- ☐ Both forward prediction and single-step retrosynthesis take only the reaction conditions and predict the yield.
- ☐ Forward prediction maps reactants to products, while single-step retrosynthesis maps a target product to candidate precursors.
- ☐ Forward prediction maps products to reactants, while single-step retrosynthesis predicts the major product from given reactants.

Question 63: Reaction templates

What role is a reaction template intended to serve?

- ☐ To give a general symbolic rule stating the required substructures and how reactants turn into products.
- ☐ To capture reaction conditions such as solvent and temperature, leaving the structural change to a neural model.

- ☐ To attach a reaction-class label grouping similar reactions, without saying how atoms or bonds change.

Question 64: Easiest reaction task

Among the tasks listed below, which one is typically the least difficult?

- ☐ Single-step retrosynthesis.
- ☐ Multi-step retrosynthesis planning.
- ☐ Forward reaction prediction.

Question 65: Template-free translation methods

In what terms do template-free translation methods formulate reaction prediction or retrosynthesis?

- ☐ As looking up the most compatible stored reaction pattern in an associative memory.
- ☐ As sorting the input into a learned reaction category and then applying a symbolic transformation.
- ☐ As a direct sequence-to-sequence translation between molecular reaction strings.

Question 66: Deep networks inside MCTS for retrosynthesis

What makes deep neural networks helpful when they are embedded within Monte Carlo Tree Search for retrosynthesis?

- ☐ They take over the search by scoring only the first disconnection and then running that route greedily.
- ☐ They can estimate the value of unvisited states and propose promising actions when exhaustive search is impractical.
- ☐ They mostly balance chemical equations at each node, while the search alone fixes value and policy without learned guidance.

Question 67: Pocket-conditioned 3D ligand generation

Which option most accurately describes pocket-conditioned 3D ligand generation?

- ☐ Ligands are built without protein context and only docked afterwards; the docking score does not steer the generator.
- ☐ The protein pocket supplies geometric context for building or denoising a ligand, usually followed by validity, docking, property, and synthesis checks.
- ☐ A fixed ligand graph is supplied and only different conformations of that same molecule are sampled, ignoring the protein.

Answer Key

Note on scoring

Questions 1–34 (the 2024 exam) are multiple-select: each listed letter is a correct option, and any option *not* listed is incorrect. On the original Moodle exam, each correct tick added points and each incorrect tick subtracted them. Questions 35–67 (the 2026 exam) are single-best-answer: exactly one letter is correct.

Q1. A, B — SMILES is the *Simplified Molecular Input Line Entry System* and is a linear character sequence. It *can* be converted to a graph, and is *not* unique (one molecule has many valid SMILES).

Q2. A, B — both `O=C(C(=O)O)C` and `CC(=O)C(=O)O` encode CH_3–CO–COOH. `CCCOOO` and `OCCOO` do not.

Q3. B, C — fingerprints are binary/numerical descriptors used for similarity search and ML. They are lossy (no unique graph reconstruction) and carry no 3D information.

Q4. A, B, C — bioassays measure a biological response, test at various concentrations (dose–response), and are wet-lab procedures. "*in vitro* = in a living organism" is false (*in vivo* is the living organism).

Q5. A, C, D — Lipinski's Ro5 is an ADME proxy, clearance is hard to model, and ADME is central to pharmacokinetics. Good ADME does *not* guarantee an effective drug.

Q6. C, D — QSAR data often comes in structurally related series and is noisy. It is usually *small* (not thousands/millions) and *imbalanced* (not balanced).

Q7. B, C — QSAR predicts drug candidates and toxicity. 3D protein modeling and genome mapping are different problems.

Q8. A — fingerprints are standard in ligand-based VS. It *is* suitable for unknown compounds, does *not* need the protein structure, and considers more than bioactivity (ADME, filters).

Q9. A, D — temporal splits simulate real-world acquisition; cluster splits keep structurally similar compounds out of both sets. Random splits do *not* eliminate bias, and CV *is* used.

Q10. B, D — BEDROC and the Enrichment Factor reward early recognition. Precision-Recall curve and Average Precision are general metrics, not early-recognition-specific.

Q11. A, B — target identification finds the *most likely* target via multi-class classification. "All proteins" and "multi-task" describe target *prediction*, not identification.

Q12. A, B — synergy is "greater than the sum"; the Loewe additivity model assesses it. Antagonism is *less* than the sum, and synergy applies to any drug pair (not only similar ones).

Q13. B, C — RF overfits less and is more interpretable than GNNs. GNNs are *not* universally better and are *not* less sensitive to data quality.

Q14. A, B — a molecular graph is a node-feature matrix plus an edge/adjacency matrix; node features include atomic number, hybridization, charge. GNNs *do* use edge information, and bond type *can* be encoded.

Q15. B, C, D — GNNs struggle with long-range interactions, very large graphs, and are compute-heavy. They are *not* "incapable of predicting any property accurately".

Q16. A, B — multi-modal contrastive learning maps modalities into a shared latent space and is *self-supervised.* It does not "concentrate representations to make predictions", and is *not* unsupervised.

Q17. C, D — a MIL bag has a *single* label and the model must be permutation-invariant. Instances are *not* individually labelled, and bag size need *not* be fixed.

Q18. D — autoregressive models can generate SMILES. Modern generators produce mostly valid SMILES, generative models explore *novel* space, and GANs *can* generate molecules.

Q19. C, D — goal-directed generation targets *specific* desired properties and can be optimised with RL. Replicating existing structures / matching the training distribution describes *distribution learning.*

Q20. B, D — FCD compares means and covariances of *latent* (ChemNet) representations and measures distance between *distributions* of generated vs. real molecules — not individual atoms, not raw structures.

Q21. A, B, C — reaction templates, training on known-reaction databases, and transformer architectures are all used. QM data is *not* required.

Q22. A, B — synthesis planning designs a reaction sequence and uses retrosynthetic analysis. It is *not* trivial, and involves more than just picking reactants.

Q23. A, C, D — per the lecture's framing the *action* is selecting reactants, the *environment* applies the reaction and returns new molecules, and the *reward* comes when the current set are all available building blocks. Only the *state* option is wrong: the state is the *current set of molecules* to be made, not the set of available building blocks.

Q24. A — cholesterol is numeric, so the *median* handles skewed missing values. Smoking status is categorical (not continuous/mean-imputable), and cholesterol is continuous, not merely ordinal.

Q25. A, C — ignored batch effects make models separate batches instead of biology, and they create artificial differences. They are *not* removed by more samples and are *not* beneficial.

Q26. C, D — changing prevalence is a label/prior shift $p(y)$; changing the diagnostic procedure is a concept shift $p(y \mid x)$. Good validation accuracy does *not* auto-adapt, and $p(x)$ *can* shift.

Q27. B, C — X-rays distinguish tissue densities; MRI uses magnetic fields and radio waves. CT uses X-rays (not ultrasound), and OCT uses light (not X-rays).

Q28. B, C — classification and segmentation are the dominant medical-imaging tasks (segmentation especially, vs. typical vision tasks).

Q29. B, D — U-Net has a symmetric encoder/decoder with skip connections (encoder feature maps concatenated into the decoder). It is convolutional (not attention-based) and was built for *segmentation*, not detection.

Q30. B — the Clever Hans effect is a model that *looks* good but relies on irrelevant features/artifacts (shortcut learning).

Q31. B, D — Grad-CAM highlights important image regions; counterfactuals show what change would flip the diagnosis. The first option mislabels LRP as "gradient integration", and occlusion perturbs *input*, not training data.

Q32. B, C — AlphaFold 2 uses multiple sequence alignments + homologous structures to predict 3D structure from sequence. It uses an attention-based (Evoformer) architecture, *not* a CNN, and no QM calculations.

Q33. A, B — proteins = amino acids via peptide bonds; DNA = nucleic acids. Sequences *do* follow rules, and RNA is nucleic acid, not lipid.

Q34. A, B, C, D — all true: motif/feature extraction, one-hot input encoding, classification/segmentation/generation tasks, and 1D-CNN / RNN / LSTM / Transformer architectures.

Q35. A — a fingerprint is computed from the graph but is lossy/hashed, so the graph cannot be recovered; "same information" (B) and "exact reconstruction" (C) are false.

Q36. B — SMILES → fingerprint is standard; fingerprints are lossy, so fingerprint → SMILES is impossible.

Q37. A — Tanimoto = shared bits divided by the union of bits (intersection over union). B is 3D pharmacophore alignment; C is scaffold matching.

Q38. A — `O=C(C)Oc1ccccc1C(=O)O` is valid acetylsalicylic acid; B misuses lowercase (aromatic) atoms for the carbonyls; C misuses square brackets.

Q39. B — bioactivity data is limited, imbalanced, and noisy, with far more inactives than actives.

Q40. C — design, make (synthesize), test, analyse, then iterate to the next design.

Q41. C — SBDD uses the target's 3D structure; LBDD uses known ligands without it.

Q42. B — similarity search retrieves neighbours of a query; model-based search uses a supervised model learned from labels.

Q43. B — AUC-ROC weights the whole ranking equally; BEDROC and the enrichment factor are early-recognition metrics.

Q44. C — use an early-recognition metric (enrichment factor near the 1% cutoff, or BEDROC).

Q45. A — split first, select features/hyperparameters on training data only, then evaluate on the held-out test set exactly once.

Q46. A — structurally related analogues split across train and test under a random split, giving an over-optimistic estimate.

Q47. A — synergy is super-additive, and a synergy-prediction model should be order-invariant (symmetric in the two compounds).

Q48. A — it jointly represents molecules *and* protein targets (single-target QSAR uses the molecule alone).

Q49. B — given a molecule and several candidate proteins, predict the most likely target.

Q50. C — node representations permute with the relabelling (equivariance), while the graph-level prediction is unchanged (invariance).

Q51. A — message generation, aggregation, update.

Q52. B — oversmoothing: node embeddings become too similar, so local atom-environment information is lost (A is over-squashing; C is too few layers).

Q53. B — pull associated molecule–target pairs together and push mismatched pairs apart.

Q54. A — a bag of feature vectors carrying a single shared label.

Q55. B — reward hacking: the generator exploited weaknesses of the proxy score rather than improving true activity.

Q56. C — always picking the single highest-scoring molecule increases exploitation and worsens rediscovery, so it is *not* a remedy (A and B both add diversity).

Q57. C — partial molecule = state, next step = action, generator = policy, completed molecule = reward.

Q58. B — diffusion starts from noise and iteratively denoises (A is one-shot decoding; C is autoregressive).

Q59. A — conditional generation: conditioned on the target property, one pass, no scoring function queried.

Q60. B — distribution learning models the training distribution and generates molecules resembling it.

Q61. B — mode collapse: the model repeatedly produces the same or very similar molecules.

Q62. B — forward prediction maps reactants → products; single-step retrosynthesis maps product → precursors.

Q63. A — a reaction template is a generalized symbolic rule giving required substructures and the reactant → product transformation.

Q64. C — forward reaction prediction is generally the easiest (single-step retro is one-to-many; multi-step planning is hardest).

Q65. C — template-free methods frame the task as direct sequence-to-sequence translation between reaction strings.

Q66. B — deep networks estimate the value of unvisited states and suggest promising actions when exhaustive search is infeasible.

Q67. B — the binding pocket provides geometric context to generate/denoise the ligand, usually followed by validity/docking/property/synthesis checks.